UNIT - II

Handwritten Study Notes | JNTU(A) R23

Syllabus Topics:

Morphometry | Soil erosion: factors, effects on fertility and land capability | Soil erosion modelling | Erosivity and erodibility | Sediment yield and sedimentation | Wetland definitions, role of water | Wetland water budgets and hydroperiod | Inflows, outflows, storage | Precipitation, runoff, evapotranspiration

1. Morphometry of a Watershed

Morphometry: the quantitative measurement and mathematical analysis of the shape, size, relief and drainage pattern of a watershed. It helps to predict hydrologic response (runoff, peak flow, erosion) even in ungauged basins.

A. Linear aspects

* Stream order (u): Strahler method - smallest unbranched streams = 1st order; when two streams of order u meet they form order u+1; a lower-order stream joining a higher order does not change the order.
* Stream number (Nu): number of streams of each order. Bifurcation ratio Rb = Nu / N(u+1). Normal range 3 - 5; high Rb (> 5) means strong structural control, elongated basin, low flood peak.
* Stream length (Lu): total length of streams of order u. Mean stream length ratio RL = Lu / L(u-1).
* Horton's laws: law of stream numbers, law of stream lengths, law of stream areas - they show geometric series relationships.
* Basin length (Lb): longest dimension of the basin parallel to main drainage line.

B. Areal aspects

Drainage density $Dd = L / A$ (Km per sq Km)

L = total length of all streams, A = basin area. High Dd: impermeable soil, sparse vegetation, steep slope -> quick runoff and high erosion. Low Dd: permeable soil, dense vegetation.

Stream frequency $Fs = N / A$ (streams per sq Km)

Form factor $Rf = A / Lb^2$

Circularity ratio $Rc = 4 \pi A / P^2$

Elongation ratio $Re = 2 \sqrt{(A/\pi)} / Lb$

Index	Low value means	High value means
Form factor Rf (max 0.78)	elongated basin, flat peak, low flood	circular basin, sharp peak, high flood
Circularity Rc (0 to 1)	elongated	circular
Elongation Re (0.6-1.0)	high relief, steep slope	low relief, flat region
Drainage density	coarse texture, permeable	fine texture, impermeable

* Constant of channel maintenance C = 1/Dd (area needed to maintain 1 Km of channel). Length of overland flow Lo = 1/(2 Dd).
* Texture ratio T = N1/P (first order streams per Km of perimeter).

C. Relief aspects

* Basin relief (H): difference in elevation between highest point and outlet.
* Relief ratio Rh = H / Lb, Ruggedness number Rn = H x Dd (high Rn: steep and long slopes, high erosion risk).
* Hypsometric curve: plot of relative area vs relative elevation; shows stage of the cycle of erosion (youth, mature, old).
* Slope: average land slope controls velocity of runoff, thus erosion and infiltration.

* Drainage patterns: dendritic (uniform rock), trellis (folded), radial (dome/volcano), rectangular (joints), parallel (steep uniform slope).

> Application:
> Morphometric analysis is used to prioritise sub-watersheds: rank those with highest Dd, Fs, Rn, Rb and lowest Rc, Rf, Re as the most erosion-prone.

2. Soil Erosion

Soil erosion: detachment, transport and deposition of soil particles by agents like water, wind, ice and gravity. Geological (normal) erosion is slow; accelerated erosion caused by human activity is the real problem. India loses about 5334 million tonnes of soil per year (about 16.4 t/ha/yr).

Types of water erosion

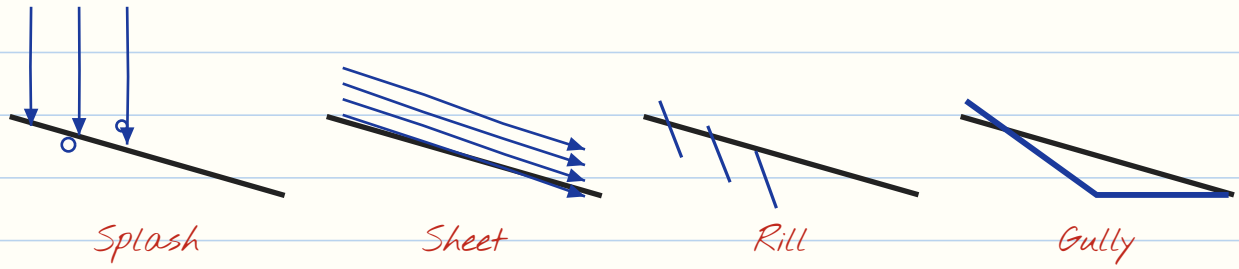


Fig 1: Stages of water erosion

* Splash erosion: raindrop impact detaches and splashes particles (first stage).
* Sheet erosion: thin uniform layer of surface soil removed by shallow runoff; hardly noticed.
* Rill erosion: small finger-like channels, < 30 cm deep, can be removed by tillage.
* Gully erosion: deep channels > 30 cm; cannot be removed by ordinary tillage; classes - small, medium, large (> 9 m).
* Stream-bank erosion, landslides/slip erosion, ravines (Chambal), sea-coast erosion.

* Wind erosion: saltation (50-75 %), surface creep (5-25 %), suspension (3-40 %). Common in arid areas.

Factors affecting erosion

Factor	How it affects erosion
Climate	rainfall amount, intensity, duration and Kinetic energy; wind speed; temperature
Soil	texture, structure, organic matter, permeability; silt and fine sand most erodible, clay and sand less
Topography	slope steepness and length; erosion increases with both
Vegetation	canopy cuts raindrop energy; roots bind soil; litter slows runoff
Land use / management	deforestation, over-grazing, up-and-down slope ploughing, shifting cultivation, mining
Human activity	construction, roads, poor drainage

> Remember:
> Soil loss increases roughly as the 1.5 to 2 power of slope length and slope steepness. Vegetative cover is the single most effective protection.

3. Effects of Erosion on Land Fertility and Capability

On-site effects

* Loss of top soil which is rich in organic matter, N, P, K and micro-nutrients -> fertility falls and yield declines (10-30 %).
* Loss of water-holding capacity and soil depth; root zone becomes shallow.
* Poor structure, crusting, low infiltration -> more runoff, more erosion (vicious circle).
* Gullies cut fields into pieces, reduce cultivable area; rocky and gravelly surface.
* Increased cost of fertiliser to compensate losses; abandonment of land.

Off-site effects

* Siltation of reservoirs, tanks and canals -> reduced storage and life of dams.
* Higher flood peaks, river bed raised, stream channel changes.
* Water pollution by sediment, nutrients and pesticides; eutrophication of lakes.
* Damage to downstream fields by sand casting, harm to aquatic life and fisheries.

Land Capability Classification (USDA, I to VIII)

Class	Suitability	Limitation / practice
I	very good cultivable land	few limits; ordinary practices
II	good cultivable	moderate limits; simple conservation (contour, rotation)
III	moderately good	severe limits, terraces, strip cropping
IV	fairly good, occasional cultivation	very severe limits; special measures
V-VII	not for cultivation: pasture, forest	steep, shallow, wet, stony
VIII	wildlife, recreation, water supply	no agricultural use

Erosion down-grades the capability class: a class II land eroded severely becomes class IV or higher. Land must be used as per capability; sub-classes e = erosion, w = wetness, s = soil, c = climate limitation.

4. Soil Erosion Modelling

Models predict soil loss, plan conservation and prioritise areas. Types: empirical (USLE), conceptual, physically-based (WEPP, EUROSEM, SWAT, AGNPS, ANSWERS).

Universal Soil Loss Equation (Wischmeier and Smith, 1978)

$$A = R\,K\,L\,S\,C\,P \quad (t/ha/yr)$$

* A = computed average annual soil loss (t/ha/yr)
* R = rainfall erosivity factor (MJ mm / ha h yr)
* K = soil erodibility factor (t ha h / ha MJ mm) - for a standard plot
* LS = slope length and steepness factor (dimensionless)
* C = cover and management factor (0 to 1, bare = 1, forest = 0.001)
* P = support (conservation) practice factor (0 to 1, no practice = 1)

Standard unit plot: 22.13 m long, 9 % slope, continuously fallow, tilled up and down slope (K measured on it).

* RUSLE (Revised USLE, 1997): improved R, K, LS, C, P using computers; MUSLE (Williams, 1975) uses runoff volume and peak rate to predict sediment yield per storm.

$$\text{MUSLE: } Y = 11.8\,(Q \times qp)^{0.56}\,K\,LS\,C\,P \quad (Y \text{ in tonnes})$$

* Limitations of USLE: estimates only sheet and rill erosion, average annual basis, no deposition, not for gullies, derived for US conditions.
* Physical models (WEPP) use infiltration, runoff routing and detachment-transport mechanics; need more data but give event and spatial results. GIS + RS used with RUSLE for large watershed maps.

Tolerance:

Soil loss tolerance (T value): maximum permissible soil loss that still allows sustained productivity: 5 to 12 t/ha/yr (India ~ 4.5 - 11.2 t/ha/yr).

5. Erosivity and Erodibility

* Erosivity: ability of rain (or wind) to cause erosion - depends on rain properties. Erosivity of a storm: EI30 = E x I30 where E = total kinetic energy of storm and I30 = maximum 30-minute intensity.

$$KE \text{ per unit rain} = 11.9 + 8.73\,\log10(i) \quad (MJ/ha/mm)$$

i = intensity in mm/h (capped at 76 mm/h). Annual R = sum of EI30 of all storms. In India simple relations like R = 79 + 0.363 P (P = annual rainfall) are used; Fournier index = p^2/P.

* Erodibility: soil's susceptibility to be detached and transported - an inherent property of the soil, not of rainfall.
* Increases with silt and very fine sand content, decreases with organic matter, clay, good structure, high permeability and aggregate stability.
* Nomograph (Wischmeier): K from % silt+very fine sand, % sand, organic matter, structure code and permeability code. Typical K: 0.02 (sandy/clay, low) to 0.07 (silty, high).
* Other indices: dispersion ratio, erosion ratio, clay ratio, aggregate stability, critical shear stress.

6. Sediment Yield and Sedimentation

Sediment yield (SY): total sediment that leaves (outflows from) a watershed measured at the outlet over a period - tonnes/Km2/year. Gross erosion minus deposition on the way.

$$SY = \text{Gross erosion} \times SDR$$

(SDR = sediment delivery ratio)

* SDR < 1: fraction of eroded soil actually reaching the outlet, falls as drainage area increases (more chance of deposition) - typical 0.05 to 0.5.
* Estimation: direct measurement (suspended sediment sampling + bed load), reservoir surveys (capacity loss), rating curve $Qs = a\,Q^b$, empirical formulas, USLE/MUSLE with SDR, models.
* Bed load (rolling, sliding, saltation) about 10-25 % and suspended load the rest, wash load = very fine, from bank/catchment.

Sedimentation in reservoirs

* Deposition of sediment in the reservoir: coarse sediment settles as delta at the head, fine sediment moves forward as turbidity currents and settles near the dam.

* Trap efficiency: percentage of inflowing sediment trapped; Brune curve gives TE as a function of capacity/inflow (C/I) ratio; large reservoirs trap > 90 %.
* Effects: loss of storage (India ~ 0.5 - 1 % per year), reduced hydropower and irrigation, downstream degradation, clogged intakes.
* Control: catchment treatment (soil conservation, check dams, afforestation), silt traps, sediment bypass, flushing / sluicing, dredging, density current venting, designing 'dead storage' for 50-100 years sediment.

Formula:

Design life of reservoir = (Dead storage) / (annual sediment inflow x trap efficiency).

7. Wetlands: Definitions and Role of Water

Wetland: land area where water is the primary factor controlling the environment and plant-animal life. Ramsar Convention (1971): areas of marsh, fen, peatland or water, natural or artificial, permanent or temporary, with water static or flowing, fresh, brackish or salt, including marine water not deeper than 6 m at low tide.

US Fish and Wildlife (Cowardin 1979): lands transitional between terrestrial and aquatic systems where the water table is usually at or near the surface or land is covered by shallow water; must have (a) hydrophytes, (b) hydric soils, (c) wetland hydrology.

Types

* Marsh (herbaceous plants, mineral soils), swamp (woody plants/trees), bog and fen (peatlands - bog gets only rain, fen groundwater too), mangroves, lagoons, estuaries, floodplain wetlands, reservoirs, rice paddies, tanks (human-made).
* India: ~ 4.6 % of area; 85+ Ramsar sites (e.g. Chilika, Keoladeo, Loktak, Sundarbans, Wular).

Role of water in wetland structure and function

* Hydrology is the master variable: depth, duration, frequency and timing of flooding decide soil chemistry (anoxic, reducing), plant species zones and animal communities.
* Water brings nutrients, sediments, seeds and removes wastes; controls oxygen, salinity, pH and temperature.
* Functions: flood storage and peak reduction, groundwater recharge/discharge, water purification (nutrient and sediment retention), shoreline stabilisation, carbon sink, habitat and biodiversity, fisheries, recreation.
* Water fluctuations make pulses of productivity; too little water -> drying and loss, too much -> drowning of vegetation.

8. Wetland Water Budget and Hydroperiod

Water budget: accounting of all water entering, leaving and stored in a wetland in a time period; it is the basis to understand hydrology and manage water level.

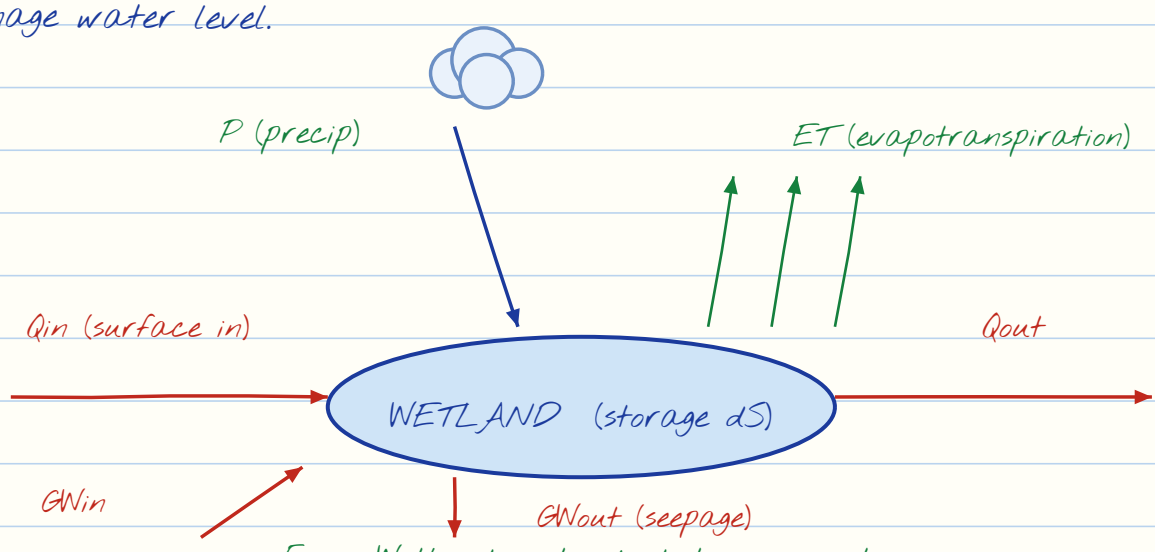


Fig 2: Wetland water budget components

$$dS/dt = P + Qsi + Gi - ET - Qso - Go$$ (storage change)

P = direct precipitation, Qsi/Qso = surface inflow/outflow (streams, overland, tidal), Gi/Go = groundwater inflow/outflow, ET = evapotranspiration, dS = change in storage (surface water + soil water).

Hydroperiod

* Hydroperiod: the seasonal pattern of water-level fluctuation in a wetland - depth, duration, frequency, timing, and rate of rise and fall of water.
* Components that control it: precipitation and evapotranspiration balance, surface water flows, groundwater exchange, tides, basin shape and outlet structure.
* Types: permanently flooded, semi-permanently flooded, seasonally flooded, saturated, temporarily flooded. Hydroperiod decides plant community and habitat for fish, amphibians and birds.
* Flooding duration is often expressed as % of year flooded, wetland criteria - saturated or flooded for >= 5 % of growing season (US) for 1 to 2 weeks.

9. Components of the Water Budget

Inflows, outflows and storage

* Inflows: precipitation, surface inflow (stream, overland sheet flow, tidal flooding), groundwater discharge (springs, seepage), human sources (irrigation return, wastewater).
* Outflows: surface outflow (stream, spillway), evapotranspiration, groundwater recharge (seepage loss), withdrawal.
* Storage: ponded surface water, soil moisture, groundwater, vegetation/biomass. Small change dS shows if wetland is gaining or losing water.

Precipitation and runoff

* Precipitation measured by rain gauges (ordinary and recording), radar, satellite, areal average by Thiessen polygon, arithmetic mean or isohyetal methods.
* Runoff entering a wetland from its catchment: Rational method $Q = C\ i\ A\ /\ 360$ (small area), SCS-CN method for volume, unit hydrograph for

peak and time.

$$\text{SCS-CN: } Q = (P - 0.2S)^2 / (P + 0.8S), \quad S = 25400/CN - 254$$

(mm)

* Curve number CN depends on soil group (A to D), land cover, treatment and antecedent moisture; higher CN = more runoff.

Evapotranspiration (ET)

* ET = evaporation from water/soil surface + transpiration through plants. Often the largest outflow in warm climates; in wetlands ET can equal or exceed Pan evaporation in summer.
* Methods: pan evaporation (ET = Kp x Epan), Penman and Penman-Monteith (energy balance + aerodynamics), Blaney-Criddle (temperature), Thornthwaite, water-budget and eddy covariance, lysimeters.

$$ETc = Kc \times ETo$$

(crop/wetland coefficient)

* ET depends on solar radiation, temperature, humidity, wind speed, vegetation type (cattail, reed), water availability and depth.

> Remember:
> Water budget of a wetland for any period: P + inflows = ET + outflows + dS. Closing the budget gives the unknown term (usually groundwater).

10. Solved Examples (for exams)

Example 1: USLE soil loss

Given: $R = 450$, $K = 0.30$, $LS = 1.2$, $C = 0.25$, $P = 0.5$. Find A.

$$A = 450 \times 0.30 \times 1.2 \times 0.25 \times 0.5 = 20.25 \text{ t/ha/yr}$$

If tolerance $T = 11$ t/ha/yr, then $A > T$ -> conservation needed. Required CP $= T / (R\,K\,LS) = 11 / (450 \times 0.3 \times 1.2) = 0.068$. So $C \times P$ must be reduced from 0.125 to about 0.068 (e.g. better cover + contour bunding).

Example 2: Drainage density

Basin area = 80 sq Km, total stream length = 200 Km, number of streams = 120, perimeter = 42 Km, basin length = 15 Km.

* Dd = 200 / 80 = 2.5 Km per sq Km (fine texture, quick runoff)
* Fs = 120 / 80 = 1.5 per sq Km
* Rf = 80 / 15^2 = 0.356 (elongated basin)
* Rc = 4 x 3.14 x 80 / 42^2 = 0.57

Example 3: Wetland water budget

Over one month: P = 120 mm, ET = 90 mm, surface inflow = 60 mm, outflow = 50 mm, groundwater net = +5 mm (in). Find dS.

$$dS = 120 + 60 + 5 - 90 - 50 = +45 \text{ mm (water level rises)}$$

> Exam tip:
> Always write the unit and the sign of each term. Units of depth (mm) over wetland area are easiest: volume = depth x area.

Quick revision list

* Morphometry: Rb, Dd, Fs, Rf, Rc, Re, Rn - linear, areal, relief.
* Erosion: splash -> sheet -> rill -> gully; USLE A = R K L S C P.
* Sediment yield = erosion x SDR; reservoir life depends on trap efficiency.
* Wetland: hydrology + hydric soil + hydrophytes; budget P + Qin + Gin = ET + Qout + Gout + dS.
* Hydroperiod = seasonal pattern of water depth - the master variable.